higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

**NATIONAL CERTIFICATE**

# LOGIC SYSTEMS N5

(8080295)

**26 November 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 6 pages.**

Copyright reserved

# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
LOGIC SYSTEMS N5
TIME: 3 HOURS
MARKS: 100

---

**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL circuits must be fully labelled.
5. Waveforms should preferably be drawn with the clock input on top followed by the LSB flip-flop output below the clock input and with the MSB flip-flop output as the last output waveform at the bottom.
6. Write neatly and legibly.

---

Copyright reserved 

## QUESTION 1

A crane control circuit requires a binary (digital) counter which generates the following output states:

$010_2$, $011_2$, $100_2$, $101_2$, $110_2$, and $111_2$.

Design a synchronous binary (digital) counter that produces the above count cycle:

The counter must be self-starting and must exit from any unused state to the starting state within ONE clock pulse.

Design and draw the circuit making use of a truth table, Karnaugh maps and THREE JK master/slave flip-flops. **[15]**

## QUESTION 2

A press safety circuit requires a binary (digital) counter which generates the following output states:

$001_2$, $010_2$, $011_2$, $100_2$, $101_2$, and $110_2$.

Draw the truth-table, circuit and the input and output waveforms for an asynchronous counter which uses gated reset/set to produce the above count cycle.

Use THREE JK master/slave flip-flops in the circuit.

The truth table and waveforms must clearly show the reset action. **[15]**

## QUESTION 3

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.1–3.13) in the ANSWER BOOK.

3.1 In a multiplexer all the data bits share the same transmission line at different times.

3.2 Multiplexing eliminates all the data transmission lines.

3.3 The purpose of multiplexing is to increase the number of transmission lines.

3.4 The purpose of multiplexing is to decrease the number of transmission lines.

3.5 A 4 to 1 multiplexer has 1 data input and 4 transmission lines.

3.6 A 4 to 1 multiplexer has 4 data inputs and 1 transmission line.

Copyright reserved 

3.7 A 1 to 4 demultiplexer has 4 data inputs and 4 output data bits.

3.8 A 1 to 4 demultiplexer has 1 data input and 4 output data bits.

3.9 Only the multiplexer requires a sequencer.

3.10 Only the demultiplexer requires a sequencer.

3.11 Both the multiplexer and the demultiplexer require a sequencer.

3.12 The multiplexer and demultiplexer need to be synchronised.

3.13 Multiplexing results in a slower transmission speed.

(13 × 1) **[13]**

**QUESTION 4**

| T | 8 | 4 | 2 | 1 |
|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 2 | 0 | 0 | 1 | 1 |
| 3 | 0 | 0 | 0 | 1 |
| 4 | 0 | 0 | 0 | 0 |
| 5 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 0 | 0 |

**FIGURE 1**

| T | 8 | 4 | 2 | 1 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 1 |
| 3 | 0 | 1 | 1 | 1 |
| 4 | 1 | 1 | 1 | 1 |
| 5 | 1 | 1 | 1 | 0 |
| 6 | 1 | 1 | 0 | 0 |

**FIGURE 2**

| T | 8 | 4 | 2 | 1 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 2 | 0 | 1 | 0 | 0 |
| 3 | 1 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 0 |
| 5 | 0 | 0 | 0 | 0 |
| 6 | 0 | 0 | 0 | 0 |

**FIGURE 3**

| T | 8 | 4 | 2 | 1 |
|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 |
| 3 | 0 | 0 | 0 | 1 |
| 4 | 1 | 0 | 0 | 0 |
| 5 | 0 | 1 | 0 | 0 |
| 6 | 0 | 0 | 1 | 0 |

**FIGURE 4**

| T | 8 | 4 | 2 | 1 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 2 | 0 | 1 | 0 | 0 |
| 3 | 1 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 1 |
| 5 | 0 | 0 | 1 | 0 |
| 6 | 0 | 1 | 0 | 0 |

**FIGURE 5**

4.1 Name the circuit which generates the truth table in FIGURE 1. (1)

4.2 Name the circuit which generates the truth table in FIGURE 2. (1)

4.3 Name the circuit which generates the truth table in FIGURE 3. (1)

4.4 Name the circuit which generates the truth table in FIGURE 4. (1)

4.5 Name the circuit which generates the truth table in FIGURE 5. (1)

4.6 What will the number be at clock pulses 9, 10 and 11 in the truth table represented in FIGURE 4? (3)

4.7 Draw the waveforms for the truth table represented in FIGURE 4. (4)
**[12]**

Copyright reserved 

## QUESTION 5

Arithmetic logic units (ALUs) perform the arithmetic operations in a computer.

Draw a fully labelled circuit diagram of a 4-bit binary adder/subtracter which uses the 2's complement method of subtraction. Indicate on the circuit ALL the logic 1's and logic 0's if the circuit has to perform 11 minus 5. **[15]**

## QUESTION 6

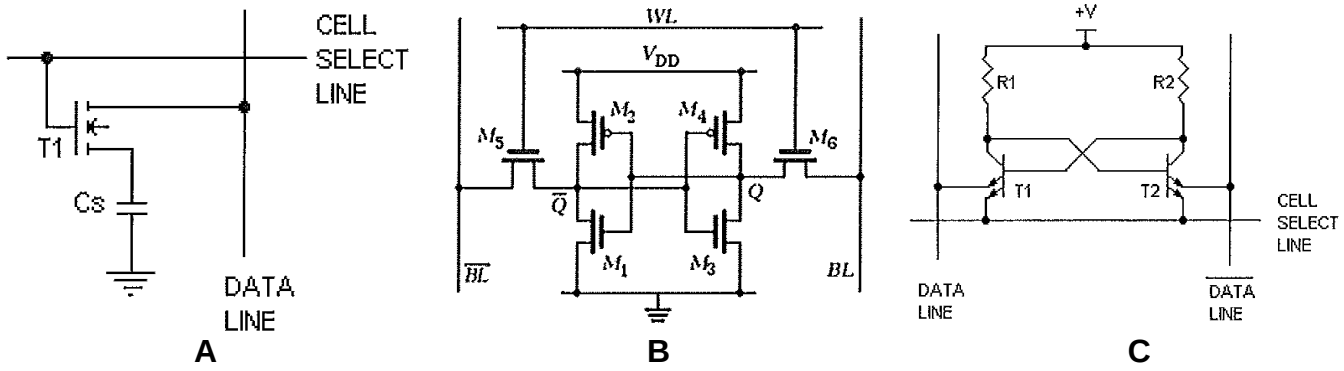



**FIGURE 6**

6.1 Refer to the three memory cells in FIGURE 6 and answer the questions.

6.1.1 Name the cell in FIGURE A.

6.1.2 Name the cell in FIGURE B.

6.1.3 Name the cell in FIGURE C.

(3 × 1) (3)

6.2 Refer to the three memory cells in FIGURE 6 and write only A, B, C or NONE next to the question number (6.2.1–6.2.7) in the ANSWER BOOK.

Which of the cells:

6.2.1 Are static cells (2)

6.2.2 Are dynamic cells (1)

6.2.3 Are volatile (3)

6.2.4 Are non-volatile (1)

6.2.5 Use UJTs (2)

6.2.6 Require refreshing circuitry (1)

6.2.7 Use cross-over feedback (2)

**[15]**

Copyright reserved 

**QUESTION 7**

Printers are output devices that are used to print documents and graphics from computers onto paper.

Explain the general operating principle of any THREE of the following printers:

7.1 Laser printer

7.2 Dot-matrix printer

7.3 Ink-jet printer

7.4 Chain printer

(Any 3 × 5) **[15]**

**TOTAL: 100**

Copyright reserved